

ALmost EXact boundary conditions for transient Schrödinger–Poisson system



Lei Bian^a, Gang Pang^b, Shaoqiang Tang^{a,*}, Anton Arnold^c

^a HEDPS, CAPT, College of Engineering, Peking University, Beijing 100871, China

^b Institute of Applied Physics and Computational Mathematics, Beijing 100088, China

^c Institut für Analysis und Scientific Computing, TU-Wien, Wiedner Hauptstr. 8, 1040 Wien, Austria

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ABSTRACT

For the Schrödinger–Poisson system, we propose an ALmost EXact (ALEX) boundary condition to treat accurately the numerical boundaries. Being local in both space and time, the ALEX boundary conditions are demonstrated to be effective in suppressing spurious numerical reflections. Together with the Crank–Nicolson scheme, we simulate a resonant tunneling diode. The algorithm produces numerical results in excellent agreement with those in Mennemann et al. [1], yet at a much reduced complexity. Primary peaks in wave function profile appear as a consequence of quantum resonance, and should be considered in selecting the cut-off wave number for numerical simulations.

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1. Introduction

The Schrödinger–Poisson equations govern many important physical systems, such as the quantum many body system, time-dependent density functional theory, and quantum charge transport in semiconductor devices [1–3].

One major challenge in numerical simulations of these equations is a broad band of wave-numbers that require well resolved to reproduce the correct physics. This leads to a huge system coupled through the electron density and self-consistent potential. When resolving each discrete wave number, we need to treat the numerical boundaries in an effective way to suppress non-physical reflections. Otherwise, wrong physics and even numerical instability will destroy the whole simulation.

In the literature, there are various treatments developed over the past two decades. An accurate approach is of DtN or NtD type [4,5]. As convolutions are involved, the boundary treatment is rather expensive numerically. Also, it does not apply to nonlinear problems in general. On the other hand, the perfectly match layer (PML) approach applies to quite general systems including the Schrödinger–Poisson system with a rather empirical choice of parameters [6,7]. More delicate approaches were developed as well [8].

In the study of accurate numerical boundary treatments for crystalline solids, we have discovered that the Bessel functions of the first kind satisfy approximate linear relations to any required accuracy [9]. This allows us to design an ALmost EXact (ALEX) boundary condition that works extremely well in that application. The time history kernel functions of the semi-discretized Schrödinger equation are also expressed in terms of the Bessel functions. Therefore, we constructed an ALEX boundary condition for the Schrödinger equation, and demonstrated its high efficiency in both linear and nonlinear

* Corresponding author.

E-mail addresses: bianlei@pku.edu.cn (L. Bian), 1517191281@qq.com (G. Pang), maotang@pku.edu.cn (S. Tang), anton.arnold@tuwien.ac.at (A. Arnold).

systems [10]. It is a boundary condition local in both space and time, with precomputed coefficients. In this work, we extend the ALEX boundary conditions to the Schrödinger-Poisson system, and demonstrate its effectiveness by applications to the resonant tunneling diode simulations. From the numerical simulations, we observe quantum resonance in the wave function profile. Two primary peaks appear inside the potential well. Their role in numerical simulation is examined. Accordingly we propose to account for the quantum resonance in selecting the cut-off wave number.

The rest of this paper is organized as follows. In Section 2, we describe the governing equations for both the stationary and the transient problem. The former is adopted for preparing the initial data to transient simulation at zero applied voltage, in order to avoid initial layer. Then in Section 3, we describe the numerical algorithms for both simulations. In particular, we provide the ALEX boundary conditions and the two-way boundary conditions. Numerical tests for a resonant tunneling diode are presented in Section 4. We explore the primary peaks in Section 5. Finally, we make some concluding remarks in Section 6.

2. Governing equations

In this section, we first describe the transient system which governs the dynamics of charge transport in a resonant tunneling diode. The initial data for this transient system are prescribed by the solution to the corresponding stationary system. The boundary conditions are also given.

2.1. Transient Schrödinger–Poisson system

In a resonant tunneling diode of length L , the electron wave function $\psi(x, k, t)$ at a wave number k is described by the Schrödinger equation

$$i\hbar\psi_t = -\frac{\hbar^2}{2m}\psi_{xx} + V(x, t)\psi, \quad x \in \mathbb{R}, \quad k \in \mathbb{R}, \quad t > 0. \quad (1)$$

Because the wave number does not appear explicitly in the equation, we may solve for each k separately. Here $\hbar$ is the reduced Planck constant, and m is the electron mass. The potential energy $V(x, t)$ consists of two parts, a self-consistent potential energy $V^s(x, t)$ and a built-in barrier potential energy $V^B(x)$ for a resonant tunneling diode. The latter is compactly supported within the device. In particular, the potential outside of the device is assumed to be

$$V(x, t) = \begin{cases} 0, & x \leq 0; \\ -eU(t), & x \geq L, \end{cases} \quad (2)$$

where $U(t)$ is a given applied voltage, and e represents the elementary charge. To avoid confusion, throughout this paper we use $\exp$ to denote the exponential function.

This potential gives rise to plane waves for the Schrödinger equation away from the space interval $[0, L]$. In particular, for $k > 0$, a plane wave takes the form of

$$\psi(x, k, t) = \begin{cases} \exp\left(-i\frac{E(k)t}{\hbar} + ikx\right) + R(k)\exp\left(-i\frac{E(k)t}{\hbar} - ikx\right), & x \leq 0; \\ T(k)\exp\left(-i\frac{E(k)t}{\hbar} + ikx\right), & x \geq L. \end{cases} \quad (3)$$

Here $E(k) = \frac{\hbar^2 k^2}{2m}$. The reflection coefficient $R(k)$ and transmission coefficient $T(k)$ are determined by the potential energy inside the device.

Similarly, for $k < 0$, a plane wave takes the form of

$$\psi(x, k, t) = \begin{cases} T(k)\exp\left(-i\frac{E(k)t}{\hbar} + ikx\right), & x \leq 0; \\ \exp\left(-i\frac{E(k)t}{\hbar} + ikx\right) + R(k)\exp\left(-i\frac{E(k)t}{\hbar} - ikx\right), & x \geq L. \end{cases} \quad (4)$$

Here $E(k) = \frac{\hbar^2 k^2}{2m} - eU(t)$. We remark that these plane waves are bounded in $L^\infty(\mathbb{R})$, not in $L^2(\mathbb{R})$.

The electron density is calculated from

$$n(x, t) = \int_{\mathbb{R}} g(k) |\psi(x, k, t)|^2 dk, \quad x \in \mathbb{R}, \quad (5)$$

with an injection profile given by the Fermi–Dirac statistics

$$g(k) = \frac{mk_B T_0}{2\pi^2 \hbar^2} \ln \left[1 + \exp \left(\frac{E_F - \hbar^2 k^2 / (2m)}{k_B T_0} \right) \right]. \quad (6)$$

Here k_B is the Boltzmann constant, T_0 is the electron temperature, and E_F is the Fermi energy. We remark that the plane waves may be regarded as injected from $\pm\infty$, hence $g(k)$ is called the injection profile.

The self-consistent potential energy $V^s(x, t)$ is coupled with the electron density by the Poisson equation in the device.

$$-\varepsilon \frac{\partial^2 V^s}{\partial x^2} = e^2 (n(x, t) - n_D(x)), \quad x \in [0, L]. \quad (7)$$

The permittivity ε and doping profile $n_D(x)$ are given constant and function, respectively. We notice that $V^s(x, t) = V(x, t)$ outside of the device, given by (2). The boundary conditions are therefore furnished with the continuity requirements at $x = 0, L$.

Instead of simulating the whole spatial domain $\mathbb{R}$, we perform numerical studies only over the device for $x \in [0, L]$, $k \in [-k_M, k_M]$ with a suitable cut-off momentum k_M , and $t > 0$. Accordingly, we need to impose suitable initial and boundary conditions for the Schrödinger equation (1).

2.2. Initial data via stationary Schrödinger–Poisson system

It has been known that quantum device simulations require well-prepared initial data to avoid strong initial layer [11, 12]. Following [1], we adopt the stationary solution $\varphi(x, k)$ as the initial data $\psi(x, k, 0)$ for $x \in [0, L]$. Furthermore, in this study we shall take $U(0) = 0$ for the transient simulation.

With an energy $E(k) = \frac{\hbar^2 k^2}{2m}$, the wave function $\varphi(x, k)$ and the stationary self-consistent potential $\bar{V}^s$ solve the following equations for each fixed $k \in \mathbb{R}$ and $x \in [0, L]$,

$$E(k)\varphi = -\frac{\hbar^2}{2m}\varphi_{xx} + [V^B(x) + \bar{V}^s(x)]\varphi, \quad (8)$$

$$-\varepsilon \frac{d^2 \bar{V}^s}{dx^2} = e^2 (\bar{n}(x) - n_D(x)). \quad (9)$$

Here $\bar{n}(x) = \int_{\mathbb{R}} g(k) |\varphi(x, k)|^2 dk$ is the electron density. The equations are solved in $[0, L]$. The boundary conditions for the Poisson equation are

$$\bar{V}^s(0) = \bar{V}^s(L) = 0. \quad (10)$$

Following [6,13], the boundary conditions for the Schrödinger equation (8) are as follows.

$$\varphi_x(0, k) + ik\varphi(0, k) = 2ik, \quad \varphi_x(L, k) = ik\varphi(L, k), \quad k > 0, \quad (11)$$

$$\varphi_x(0, k) = ik\varphi(0, k), \quad \varphi_x(L, k) + ik\varphi(L, k) = 2ike^{ikL}, \quad k < 0. \quad (12)$$

2.3. Boundary conditions

On the boundaries of the device, incoming waves enter from $x = 0$ for $k > 0$, and from $x = L$ for $k < 0$.

For $k > 0$, we notice that $\exp(-i\frac{E(k)}{\hbar}t + ikx)$ for $x \leq 0$ represents an incoming wave for the left boundary $x = 0$. Therefore, $\psi(x, k, t) - \exp(-i\frac{E(k)}{\hbar}t + ikx) = R(k) \exp(-i\frac{E(k)}{\hbar}t - ikx)$ satisfies a transparent boundary condition for outgoing waves at $x = 0$, and $\psi(x, k, t) = T(k) \exp(-i\frac{E(k)}{\hbar}t + ikx)$ satisfies a transparent boundary condition for outgoing waves at $x = L$. We remark that depending on the formulation of transparent boundary condition, one should be careful of consistency at $t = 0$ [14].

In the same way, for $k < 0$, $\psi(x, k, t)$ at $x = 0$ and $\psi(x, k, t) - \exp(-i\frac{E(k)}{\hbar}t + ikx)$ at $x = L$ satisfy transparent boundary conditions for outgoing waves, respectively.

We remark that in the literature, there are only several numerically effective ways in imposing such transparent boundary conditions, in spite of considerable efforts put in this important yet difficult topic [7,15–18].

3. Numerical methods

In this section, we first briefly describe the Gummel iteration method for solving the stationary Schrödinger–Poisson system [1]. Then we present the discretization and boundary treatment for the transient problem.

3.1. Stationary Schrödinger–Poisson system

We discretize the space x and the momentum k with equidistant grids, respectively. Let $x_j = j\Delta x$, for $j = 0, \dots, J$ with $x_J = L$; and $k_l = l\Delta k$, for $l = -M, \dots, -1, 1, \dots, M$, with $k_M = \sqrt{2m(E_F + 7k_B T_0)}/\hbar$ [11].

The equations (8) and (9) are iteratively solved for each $k = k_l$ as follows, with p counting the iteration step,

$$E^d(k)\varphi_{j,l}^{(p)} = -\frac{\hbar^2}{2m(\Delta x)^2}(\varphi_{j+1,l}^{(p)} - 2\varphi_{j,l}^{(p)} + \varphi_{j-1,l}^{(p)}) + [V^B(x_j) + \bar{V}_j^p]\varphi_{j,l}^{(p)}, \quad (13)$$

$$-\frac{\varepsilon}{(\Delta x)^2}(\bar{V}_{j-1}^{(p+1)} - 2\bar{V}_j^{(p+1)} + \bar{V}_{j+1}^{(p+1)}) = e^2 \left[\bar{n}_j^{(p)} \exp\left(\frac{\bar{V}_j^{(p)} - \bar{V}_j^{(p+1)}}{V^{ref}}\right) - n_D(x_j) \right]. \quad (14)$$

Here $E^d(k) = \frac{\hbar^2}{m(\Delta x)^2}(1 - \cos(k\Delta x))$ is the dispersion relation for the second order finite difference, $V^{ref} = 0.03$ eV is a relaxation parameter, and the electron density is calculated from

$$\bar{n}_j^{(p)} = \Delta k \sum_{|l|=1}^M g(k_l) |\varphi_{j,l}^{(p)}|^2. \quad (15)$$

Following [1,14], we take boundary conditions for (13) as follows,

$$-\frac{1}{\alpha^+}\varphi_{-1,l}^{(p)} + \varphi_{0,l}^{(p)} = 1 - \frac{1}{(\alpha^+)^2}, \quad \varphi_{J,l}^{(p)} - \frac{1}{\alpha^+}\varphi_{J+1,l}^{(p)} = 0, \quad k > 0, \quad (16)$$

$$\varphi_{0,l}^{(p)} - \alpha^-\varphi_{-1,l}^{(p)} = 0, \quad \varphi_{J,l}^{(p)} - \frac{1}{\alpha^-}\varphi_{J+1,l}^{(p)} = \left(\frac{1}{\alpha^-}\right)^J - \left(\frac{1}{\alpha^-}\right)^{J+2}, \quad k < 0, \quad (17)$$

with

$$\alpha^\pm = 1 - \frac{mE^d(k)(\Delta x)^2}{\hbar^2} \pm i\sqrt{\frac{2mE^d(k)(\Delta x)^2}{\hbar^2} - \frac{m^2(E^d(k))^2(\Delta x)^4}{\hbar^4}}. \quad (18)$$

The boundary conditions for (14) are homogeneous $\bar{V}_0^{(p+1)} = \bar{V}_J^{(p+1)} = 0$.

3.2. Transient Schrödinger–Poisson system

For the transient problem, the discretized Poisson equation is

$$-\frac{\varepsilon}{(\Delta x)^2}(V_{j-1}^s - 2V_j^s + V_{j+1}^s) = e^2(n_j(t) - n_D(x_j)), \quad j = 1, \dots, J-1, \quad (19)$$

with $V_0^s = 0$ and $V_J^s = -eU(t)$. Here the electron density is calculated from

$$n_j(t) = \Delta k \sum_{|l|=1}^M g(k_l) |\psi_{j,l}(t)|^2. \quad (20)$$

The Schrödinger equation is semi-discretized in space as

$$i\hbar \frac{d}{dt}\psi_{j,l} = -\frac{\hbar^2}{2m(\Delta x)^2}(\psi_{j+1,l} - 2\psi_{j,l} + \psi_{j-1,l}) + V_j\psi_{j,l}, \quad j \in \mathbb{Z}, \quad (21)$$

with $V_j = V_j^s + V^B(x_j)$.

In the previous work [10], we have discovered that the semi-discretized system (21) admits time history kernel function (lattice Green's function) in the form of Bessel functions. That is, as $V_j = 0$ for $j \leq 0$ in (21), we have

$$\psi_{-p,l}(t) = \frac{p}{t} i^p \exp(-2it) J_p\left(\frac{\hbar t}{m(\Delta x)^2}\right) * \psi_{0,l}(t). \quad (22)$$

Meanwhile, through an asymptotic expansion it was discovered that the first kind Bessel functions are approximately linearly dependent to any required accuracy, provided their orders are big enough [9]. More precisely, for $M_p \geq 5$ and any given $\varepsilon > 0$, if Q is big enough, there exist a set of real numbers a_p ($1 \leq p \leq M_p$), such that

$$\left| \sum_{p=1}^{M_p} a_p J_{Q-p}(t) - J_Q(t) \right| < \frac{\varepsilon}{\sqrt[3]{Q}}. \quad (23)$$

This naturally yields a transparent boundary condition

$$\psi_{-Q,l}(t) \approx \sum_{p=1}^{M_p} a_p \psi_{-Q+p,l}(t). \quad (24)$$

Using the recursive relation for the derivative of the Bessel functions, we may further enhance the efficiency for numerical boundary treatments. After some calculations, the approximate linear dependency leads to an ALmost EXact (ALEX) boundary condition as follows.

$$i\hbar \frac{d}{dt} \psi_{-Q,l} = \sum_{p=0}^{M_p} b_p \psi_{-Q+p,l}. \quad (25)$$

In this study, we take the same formulation as [10] with $M_p = 15$ and b_0 to b_{15} as follows.

$$\begin{aligned} & -\hbar^2 / (2m(\Delta x)^2) \times \\ & -2 + 12.052110193235i, \quad 71.177637000065, \\ & 262.693855125265i, \quad 708.598614518240, \\ & 1461.844046849774i, \quad 2387.382683759241, \\ & 3149.979100974855i, \quad 3395.091578678526, \\ & 3000.639882589720i, \quad 2168.648792241722, \\ & 1268.957016689205i, \quad 589.685213345064, \\ & 210.491232766998i, \quad 54.442223074977, \\ & 9.111658309311i, \quad 0.742160906401. \end{aligned}$$

Now we present the boundary conditions based on ALEX [19].

First, we consider the left boundary $x = 0$. For $k < 0$, we directly take the boundary condition

$$i\hbar \frac{d}{dt} \psi_{0,l} = \sum_{p=0}^{M_p} b_p \psi_{p,l}. \quad (26)$$

For $k > 0$, we adopt the ALEX boundary condition for the reflected component $\psi_{j,l} - \exp(i(kj\Delta x - \frac{E^d(k_l)}{\hbar}t))$. This gives

$$i\hbar \frac{d}{dt} \psi_{0,l} = \sum_{p=0}^{M_p} b_p \psi_{p,l} + \exp\left(-i\frac{E^d(k_l)}{\hbar}t\right) E^d(k_l) - \exp\left(-i\frac{E^d(k_l)}{\hbar}t\right) \sum_{p=0}^{M_p} b_p \exp(ikp\Delta x). \quad (27)$$

Next, we consider the right boundary $x = L$. Using a transform $\tilde{\psi}_{j,l} = \psi_{j,l} \exp(-\frac{ie}{\hbar} \int_0^t U(s)ds)$, one may show that $\tilde{\psi}_{j,l}$ is governed by the free discrete Schrödinger equation, namely, without the electric potential term. For $k < 0$, we adopt the ALEX boundary condition for the reflected part $\tilde{\psi}_{j,l} - \exp(i(kj\Delta x - \frac{E^d(k_l)}{\hbar}t))$. Going back to $\psi_{j,l} = \tilde{\psi}_{j,l} \exp(\frac{ie}{\hbar} \int_0^t U(s)ds)$, we get the boundary condition

$$i\hbar \frac{d}{dt} \psi_{J,l} = \sum_{p=0}^{M_p} b_p \psi_{J-p,l} + V_J \psi_{J,l} + \exp\left(\frac{ie}{\hbar} \int_0^t U(s)ds - i\frac{E^d(k_l)}{\hbar}t\right) \left[E^d(k_l) \exp(ikL) - \sum_{p=0}^{M_p} b_p \exp(ik(L - p\Delta x)) \right]. \quad (28)$$

For $k > 0$, the ALEX boundary condition for $\tilde{\psi}_{j,l}$ leads to

$$i\hbar \frac{d}{dt} \psi_{J,l} = \sum_{p=0}^{M_p} b_p \psi_{J-p,l} + V_J \psi_{J,l}. \quad (29)$$

3.3. Algorithm for the transient computation

We summarize one-step advancing for the transient computation as follows.

Let the wave function at a step $t^r = r\Delta t$ be $\psi_{j,l}^r$ for $j = 0, \dots, J$ and $|l| = 1, \dots, M$.

Table 1
Physical constants and parameters in this simulation.

Symbol	Physical meaning	Value
e	Elementary charge	1.602×10^{-19} C
$\hbar$	Reduced Planck constant	1.0546×10^{-34} J · s
m	Effective mass	6.1033×10^{-32} kg
ε	Permittivity	1.012319×10^{-10} F/m
k_B	Boltzmann constant	$1.3806505 \times 10^{-23}$ J/K
E_F	Fermi energy	6.7097×10^{-21} J
T_0	Semiconductor temperature	300 K

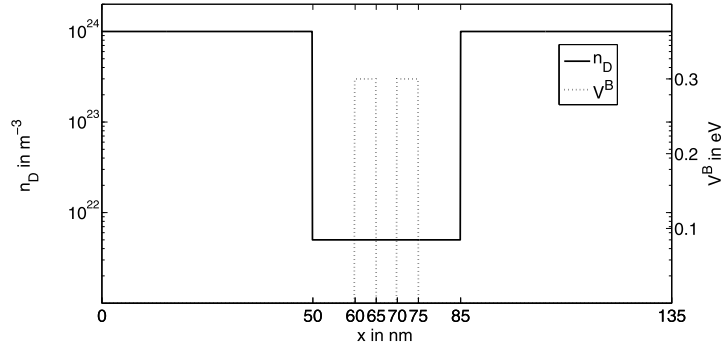


Fig. 1. The doping profile and the given barrier potential energy.

- Step 1 Evaluate the electron density n_j^r according to (20) for $j = 1, \dots, J - 1$;
 Step 2 Solve the Poisson equation (19) by a standard solver to obtain the self-consistent electron potential energy and hence the total potential energy V_j^r for $j = 1, \dots, J - 1$;
 Step 3 Under potential energy V_j^r , use the Crank–Nicolson scheme to solve the ordinary differential equation (21) at $j = 1, \dots, J - 1$ with the ALEX boundary conditions (26)–(29).

3.4. Implementation details

The physical constants used in this study are listed in Table 1.
 The electron doping profile is

$$n_D(x) = \begin{cases} 1 \times 10^{24} \text{ m}^{-3}, & x \in [0, 50 \text{ nm}] \cup [85 \text{ nm}, 135 \text{ nm}], \\ 5 \times 10^{21} \text{ m}^{-3}, & \text{elsewhere.} \end{cases} \quad (30)$$

The barrier potential is

$$V^B(x) = \begin{cases} 0.3 \text{ eV}, & x \in [60 \text{ nm}, 65 \text{ nm}] \cup [70 \text{ nm}, 75 \text{ nm}], \\ 0, & \text{elsewhere.} \end{cases} \quad (31)$$

They are schematically shown in Fig. 1.

We take $J = 1350$ and $M = 1500$. We note that $k_M = 6.26 \times 10^8 \text{ m}^{-1}$, according to the choice of physical constants and parameters.

4. Numerical test

We first present the stationary simulation results at zero applied voltage. Then we show the transient simulation results with initial data given by the stationary solution.

4.1. Numerical solution of stationary Schrödinger–Poisson system

For the stationary solution to the Schrödinger–Poisson system at zero applied voltage, Fig. 2 shows $|\varphi(x, k)|$ in the (x, k) -plane. A pair of peaks is clearly visible. They are located around $k \approx \pm 1174 \Delta k$, and within the barrier region [60 nm, 75 nm]. See Fig. 3. The peak's amplitude is 25.5385, much higher than elsewhere. They correspond to electrons trapped in the barrier, see Fig. 4. We call these peaks the *primary peaks*.

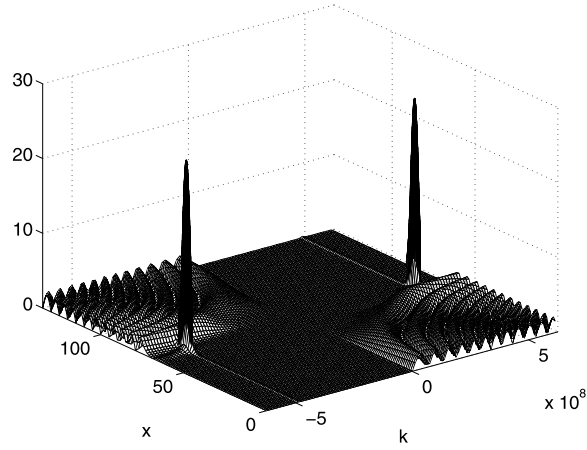


Fig. 2. Wave function $|\varphi(x, k)|$ in the (x, k) -plane.

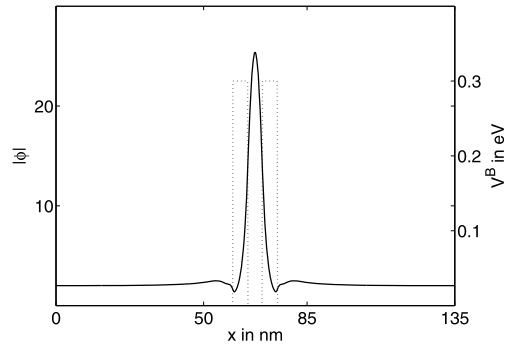


Fig. 3. Wave function $|\varphi(x, k)|$: a “side view” ($\max_k |\varphi(x, k)|$).

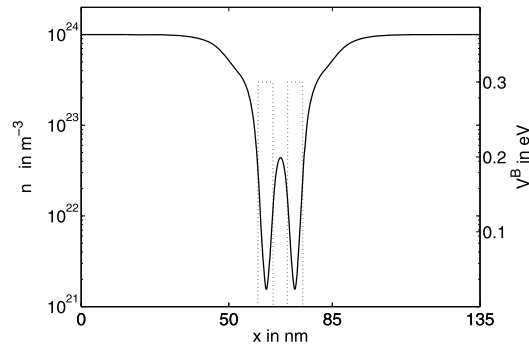


Fig. 4. The electron density $\bar{n}(x)$.

The conduction current density

$$J_{cond}(x) = \frac{e\hbar}{m} \int_{-k_M}^{k_M} g(k) \text{Im} \left(\varphi^*(x, k) \frac{\partial \varphi(x, k)}{\partial x} \right) dk, \quad (32)$$

is depicted in Fig. 5. Noticing that the current density is on the order of 10^9 A/m² when there is an applied voltage, this current density is practically regarded as 0, as is expected for zero applied voltage.

The self-consistent potential energy $\bar{V}^s$ is shown in Fig. 6. The doping profile, which is low for the depleted region [50 nm, 85 nm], as well as a minor effect from the electron density variation due to the double barrier.

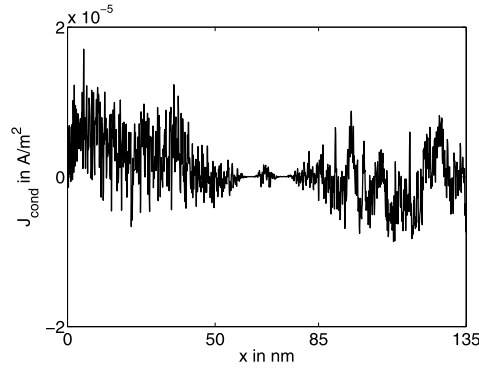


Fig. 5. Current density in the stationary Schrödinger–Poisson system.

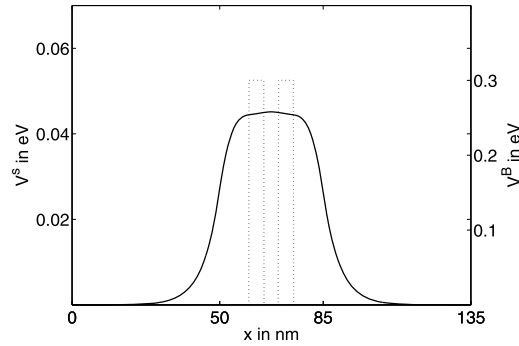


Fig. 6. Self-consistent potential energy in the stationary Schrödinger–Poisson system.

4.2. Numerical solution of transient Schrödinger–Poisson system

We solve the transient system with the ALEX boundary condition and the Crank–Nicolson scheme. The time step is chosen as $\Delta t = 0.5$ fs. The applied voltage is

$$U(t) = \begin{cases} \frac{U_0}{2} (1 - \cos \frac{2\pi t}{T}), & 0 \leq t \leq 1 \text{ ps}; \\ U_0, & t > 1 \text{ ps}, \end{cases} \quad (33)$$

where $U_0 = 0.25$ V and $T = 2$ ps.

First, we demonstrate that the proposed boundary treatments effectively suppress spurious reflections. In Fig. 7, we show the wave function modulus $|\psi(x, k, 0.2 \text{ ps})|$. As mentioned before, at the left boundary of the device ($x = 0$) there is no incoming wave for $k < 0$. The numerical solution does not show any spurious reflection. Similarly, no reflection is observed at the right boundary ($x = 135$ nm) for $k > 0$. Moreover, the primary peaks remain at the same (x, k) -position, compared to the zero-bias case at $t = 0$, see Fig. 2. Yet their amplitudes decrease slightly to 25.3392. The change in wave function modulus is shown in Fig. 8. The biggest change occurs along $\pm 1174 \Delta k$.

To check the effectiveness of the boundary treatment in more detail, we focus on a particular wave number $k = \Delta k$, and plot the evolution of the wave function modulus $|\psi(x, k, t)|$ for $t \in [0, 0.2 \text{ ps}]$ in the upper-left subplot of Fig. 9. In the upper-right plot, we zoom in the right-boundary vicinity $x \in [130 \text{ nm}, 135 \text{ nm}]$. The directions of contour lines clearly demonstrate that the wave propagates from left to right as time elapses, without observable reflections. For the left boundary vicinity $x \in [0 \text{ nm}, 5 \text{ nm}]$ in the lower-left subplot, the wave function consists of both an incoming part, and the rest that comes from wave propagation and complex dynamics. In the lower-right subplot, we show that the mentioned rest is indeed an outgoing wave that propagates from right to left.

With the numerical boundary treatment validated, we now briefly describe the behavior of the device, which agrees with those in the literature [10–12]. As seen from Fig. 7, the peaks persist over the whole simulation, yet the amplitude varies. A close check shows that the profiles at other wave numbers and positions also change along with time. The evolution in the wave function leads to change in the electron density, current and self-consistent potential.

From Fig. 10, we observe that, when the applied voltage increases from 0 to $\frac{U_0}{2}$ at $t = 0.5$ ps, the electron density increases in the left barrier and the intermediate region between the two barriers, which clearly reflects the resonant tunneling mechanism. The electron density inside the right barrier does not change much, and to the right of it, it decreases considerably. Accordingly, the conduction current increases in this period. Afterwards, when the applied voltage further

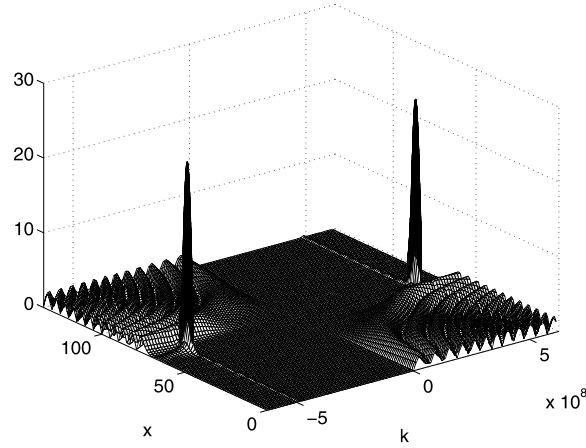


Fig. 7. Wave function $|\psi(x, k, 0.2 \text{ ps})|$.

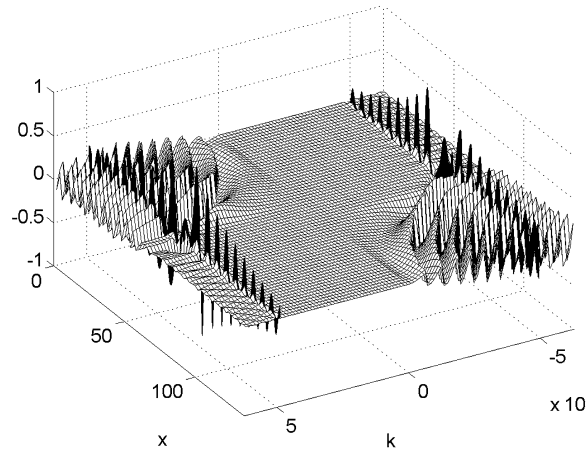


Fig. 8. Change in wave function modulus $|\psi(x, k, 0.2 \text{ ps})| - |\psi(x, k, 0 \text{ ps})|$.

increases up to U_0 at $t = 1 \text{ ps}$, the electron density moves up in the left barrier and the intermediate region. The conduction current density also moves up accordingly. However, at the left doping gradient boundary, a minor kink develops at about $x = 52 \text{ nm}$. This may mean that there are not enough electrons from the right side, yet the increment in applied voltage attracts more electrons to the left. The conduction current density moves downward away from the barriers, leading to a smaller total current. After $t = 1 \text{ ps}$, when the applied voltage remains at U_0 , there are more electrons accumulated in the intermediate region, and the conduction current density profile becomes flatter in the device.

In Fig. 11, we plot the total current density averaged over $[0, L]$ during the whole time period, which is defined as

$$J_{\text{total}} = J_{\text{cond}} - \frac{\varepsilon}{e} \frac{\partial}{\partial t} \nabla V^s. \quad (34)$$

This agrees with [1]. If we take the discrete transparent boundary conditions with the exact convolution from [14,1] as the reference solution, we observe that the numerical error in our solution is negligible, cf. Fig. 12. Λ is the parameter which appeared in the fast evaluation for the discrete convolution terms. Defining the relative L^2 -error as

$$\varepsilon_r(t) = \frac{\sqrt{\int_0^L (J_{\text{cond}}(x, t) - J_{\text{cond}}^{\text{ref}}(x, t))^2 dx}}{\sqrt{\int_0^L (J_{\text{cond}}^{\text{ref}}(x, t))^2 dx}}, \quad (35)$$

we obtain $\varepsilon_r(1.5 \text{ ps}) = 6.45 \times 10^{-10}$. This is much smaller than that for $\Lambda = 100$ in the fast evaluation [1], and demonstrates the effectiveness of our boundary treatment.

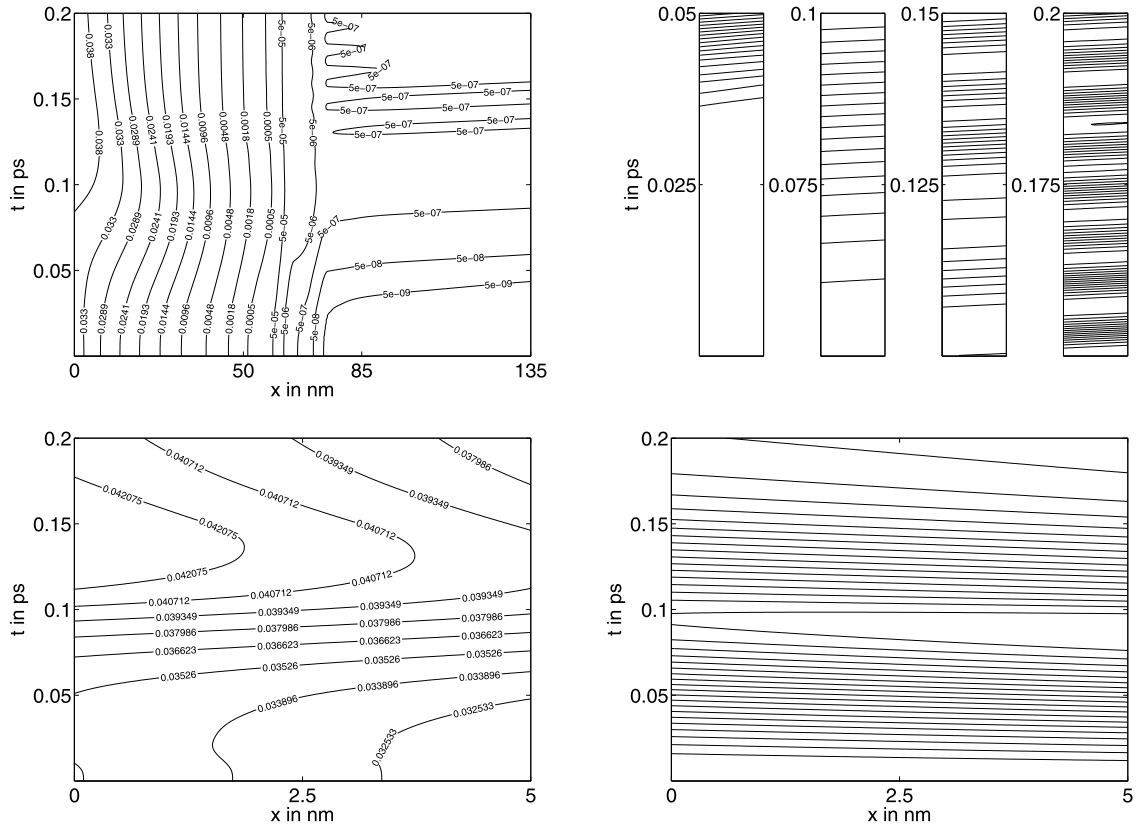


Fig. 9. $|\psi(x, \Delta k, t)|$ for $t \in [0, 0.2]$ ps. Upper-left: wave function in the whole device; upper-right: blow-up view near the right boundary for $x \in [130 \text{ nm}, 135 \text{ nm}]$, time $t \in [0, 0.2]$ ps in four equal length t -intervals; lower-left: blow-up view near the left boundary for $x \in [0, 5 \text{ nm}]$; lower-right: $|\psi(x, \Delta k, t) - \exp(-iE^d(\Delta k)/\hbar t + i\Delta kx)|$ near the left boundary for $x \in [0, 5 \text{ nm}]$.

5. Primary peaks

The primary peaks are prominent in the wave function profile of a resonant tunneling diode. They are induced by the double barrier structure, and influence the physics in the device.

To better understand these peaks, we first double the computational domain in wave number space, namely, $k \in [-3000\Delta k, 3000\Delta k]$. As shown in Fig. 13 for the stationary Schrödinger–Poisson simulation, a pair of secondary peaks appears at $k \approx \pm 1866\Delta k$, with a much smaller amplitude. Because of the fast decay in the Fermi–Dirac distribution (6), the extended computational domain, and in particular the secondary peaks do not lead to a significant change in the electron density n or the potential energy V . Moreover, it is observed that the secondary peaks are at an energy level equal to the barrier potential, namely, $\frac{\hbar^2 k^2}{2m} = e(V^B + \bar{V}^S) = 5.5269 \times 10^{-20}$ J.

To further explore the choice of the wave number domain, we calculate the electron density $\bar{n}_j$ with different cut-off wave numbers. In analogy to the partial sum in (15), we consider now $\bar{n}_j^M \equiv \Delta k \sum_{|l|=1}^M g(k_l) |\varphi_{j,l}|^2$. Fig. 14 shows $\bar{n}_j^M$ as a function of M . For the positions $x \leq 60 \text{ nm}$ (and by symmetry also for $x \geq 75 \text{ nm}$), the function $\bar{n}_j^M$ almost saturates for $M = 800$. But for the positions $62.5 \text{ nm} \leq x \leq 67.5 \text{ nm}$, $\bar{n}_j^M$ has a significant jump close to $M = 1174$. This explains that the choice of $k_M = \sqrt{2m(E_F + 7k_B T_0)/\hbar}$ is sufficient in (6) for 0 applied voltage. In Fig. 4, we plotted the electron density for summation over the whole computational domain (with respect to the wave number). The hump inside the barrier is caused by the two peaks, as explained above.

The primary peaks may be explained as a resonance phenomenon in quantum device. In a much simplified manner, we may consider the following stationary Schrödinger equation

$$E(k)\phi = -\frac{\hbar^2}{2m}\phi_{xx} + V(x)\phi, \quad x \in \mathbb{R}, \quad (36)$$

under potential

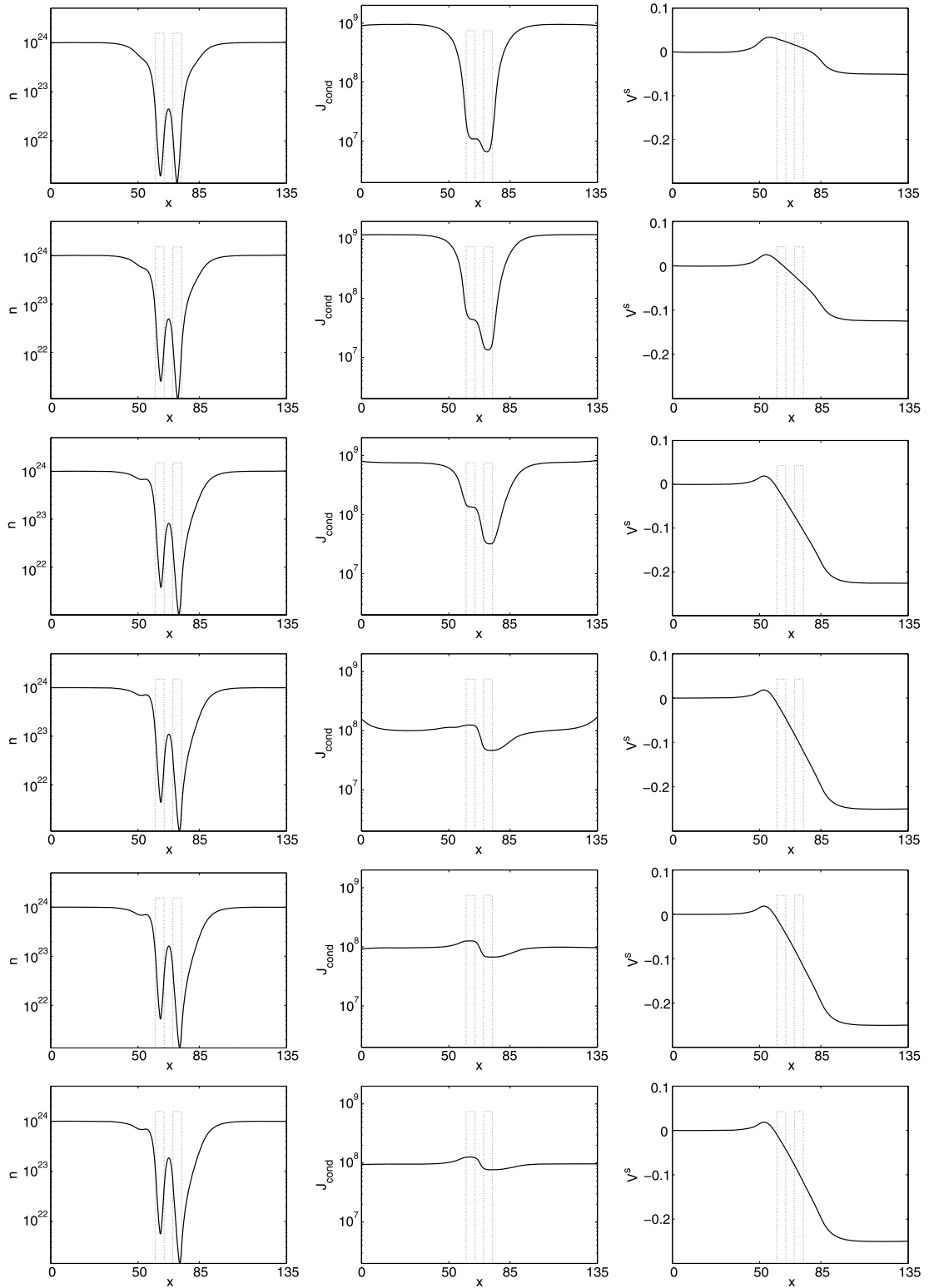


Fig. 10. Left column: electron density in m^{-3} ; middle column: current density in A/m^2 ; right column: self-consistent potential energy in eV. Different rows show different times: $t = 0.3$ ps, 0.5 ps, 0.8 ps, 1.0 ps, 1.5 ps and 2.0 ps (top to bottom), respectively.

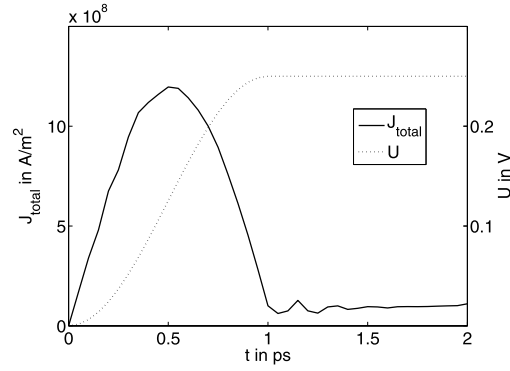


Fig. 11. Solid line: total current density as a function of t , averaged in space. Dotted line: applied bias as a function of t .

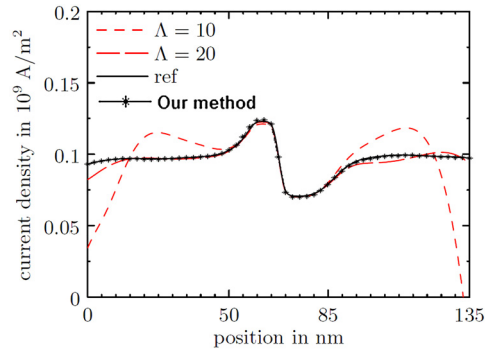


Fig. 12. Conduction current density at $t = 1.5$ ps with comparison to [1].

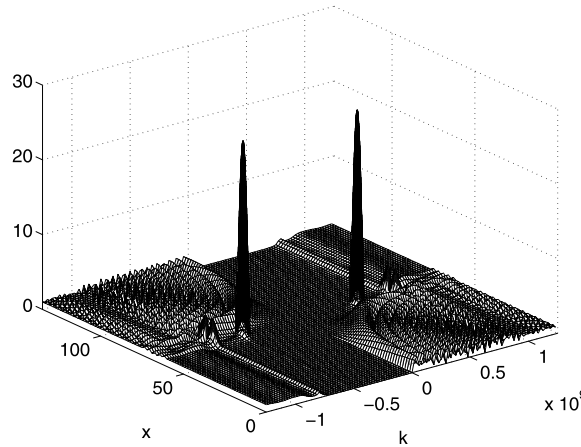


Fig. 13. Wave function $|\varphi(x, k)|$ computed with cut-off wave number $k'_M = 2k_M$.

$$V(x) = \begin{cases} 0, & x < 60 \text{ nm}; \\ 0.345 \text{ eV}, & 60 \text{ nm} < x < 65 \text{ nm}; \\ 0.045 \text{ eV}, & 65 \text{ nm} < x < 70 \text{ nm}; \\ 0.345 \text{ eV}, & 70 \text{ nm} < x < 75 \text{ nm}; \\ 0, & x > 75 \text{ nm}. \end{cases} \quad (37)$$

We rescale $V(x)$ by 0.345 eV, shift x by 67.5 nm and rescale it by $\hbar/\sqrt{2m * 0.345 \text{ eV}}$, and rescale k by $\sqrt{2m * 0.345 \text{ eV}}/\hbar$. Let $w = 2.5 \text{ nm}/[\hbar/\sqrt{2m * 0.345 \text{ eV}}] = 1.9471096$, $q = 0.045 \text{ eV}/0.345 \text{ eV} = 0.1304348$. Then the rescaled equation and potential are as follows:

$$E(k)\phi = -\phi_{xx} + V(x)\phi, \quad x \in \mathbb{R}, \quad (38)$$

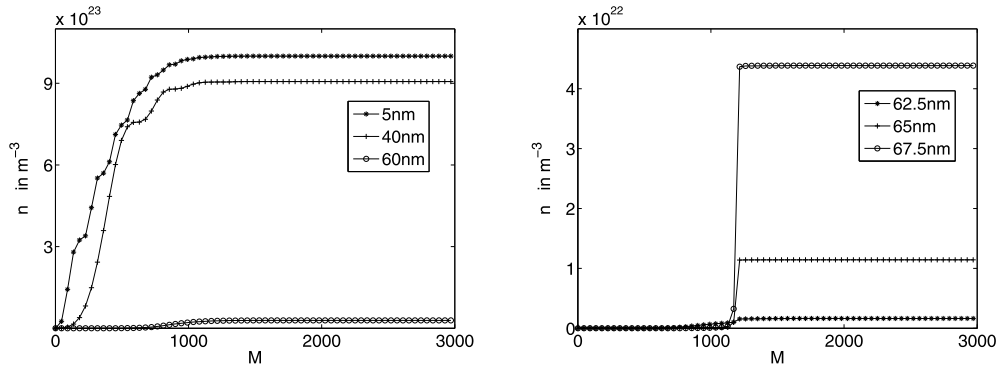


Fig. 14. Electron density computed with different number of terms M .

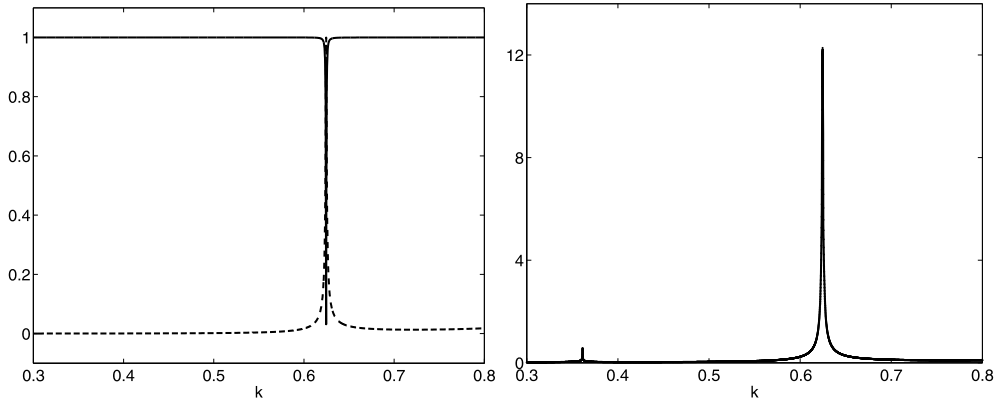


Fig. 15. Resonance computed with the simplified model. Left: reflection coefficient $|R(k)|$ (solid line) and transmission coefficient $|T(k)|$ (dashed line) as function of k . Right: matching coefficients $|A_0|$ (solid line) and $|B_0|$ (dashed line) from (40), which coincide with each other. They represent the amplitude of the two traveling waves inside the potential barriers. The horizontal axis stands for the undimensionalized wave number k .

$$V(x) = \begin{cases} 0, & x < -3w; \\ 1, & -3w < x < -w; \\ q, & -w < x < w; \\ 1, & w < x < 3w; \\ 0, & x > 3w. \end{cases} \quad (39)$$

Further let $\alpha = \sqrt{1 - k^2}$ and $\beta = \sqrt{k^2 - q}$. The analytical solution with an incident wave $\exp(ikx)$ takes the following form

$$\phi(x, k) = \begin{cases} \exp(ikx) + R(k) \exp(-ikx), & x < -3w; \\ A_- \exp(\alpha x) + B_- \exp(-\alpha x), & -3w < x < -w; \\ A_0 \exp(i\beta x) + B_0 \exp(-i\beta x), & -w < x < w; \\ A_+ \exp(\alpha x) + B_+ \exp(-\alpha x), & w < x < 3w; \\ T(k) \exp(ikx), & x > 3w. \end{cases} \quad (40)$$

Here the coefficients A_- , A_0 , A_+ , B_- , B_0 , B_+ are to be determined, $R(k)$ and $T(k)$ are reflection and transmission coefficients, respectively. Making use of $\phi(x) \in C^1$, we may form a linear algebraic system for the above coefficients. Eliminating the matching coefficients, we end up with

$$\begin{bmatrix} T(k) \exp(3ikw) \\ 0 \end{bmatrix} = -\frac{1}{16\alpha^2\beta k} \begin{bmatrix} \alpha + ik & -\alpha + ik \\ -\alpha + ik & \alpha + ik \end{bmatrix} \Lambda(2\alpha) \begin{bmatrix} \alpha + i\beta & \alpha - i\beta \\ \alpha - i\beta & \alpha + i\beta \end{bmatrix} \Lambda(2i\beta) \\ \cdot \begin{bmatrix} \alpha + i\beta & -\alpha + i\beta \\ -\alpha + i\beta & \alpha + i\beta \end{bmatrix} \Lambda(2\alpha) \begin{bmatrix} \alpha + ik & \alpha - ik \\ \alpha - ik & \alpha + ik \end{bmatrix} \Lambda(-3ik) \begin{bmatrix} 1 \\ R(k) \end{bmatrix}, \quad (41)$$

with the diagonal matrix

$$\Lambda(s) = \begin{bmatrix} \exp(sw) & \\ & \exp(-sw) \end{bmatrix}.$$

Solving this linear system for R and T , we find that resonance, i.e., $|T(k)| = 1$ occurs at about $k^* = 0.62458$. See Fig. 15. This gives, in physical units, a wave number of $0.62458 \times \sqrt{2m \cdot 0.345 \text{ eV}/\hbar} = 4.8645 \times 10^8 \text{ m}^{-1}$. In our stationary solution, the primary peaks appear at $1174\Delta k = 4.8995 \times 10^8 \text{ m}^{-1}$. Hence, the relative error from the simplified linear model (38)–(39) is less than 1%. Numerically, the transmission and reflection coefficients are $|T(k^*)| = 0.999972$, $|R(k^*)| = 0.007418$. Moreover, the maximal amplitude of $|\phi(x, k^*)| = 24.5175$ at $x = 0$. Again it agrees well with the full stationary Schrödinger–Poisson simulation, which gives 25.5385.

With the above discussions, we propose to prescribe the cut-off wave number in a new way, namely, $\tilde{k}_M \equiv \max(\sqrt{2m(E_F + 7k_B T_0)/\hbar}, \sqrt{2m \cdot \max V^B(x)/\hbar})$, where the second term corresponds to the level of the potential barrier. Under the settings described before, this amounts to $7.26 \times 10^8 \text{ m}^{-1}$.

6. Conclusion

In this paper, we propose the ALEX boundary treatment to effectively suppress spurious reflections in the wave functions at the ends of the resonant-tunneling diode device. It yields high accuracy, comparable to the theoretically exact boundary treatment developed in [15] and applied in [1]. The numerical cost and programming complexity are reduced considerably. Being local in both space and time, it bears nice features of being simple, clear, and without empirical parameters. When the mesh size or time step size changes during a simulation, it is not necessary to restart the simulation from the beginning, in contrast to the time history dependent boundary treatments in [1]. The numerical results with the proposed ALEX boundary treatments demonstrate the effectiveness.

In the wave function profile, we observe two primary peaks around $\pm 1174\Delta k$, located inside the barrier potential well in space. With the choice of cut-off wave number proposed in previous works [20,1], these peaks are included in the computational domain. The important role of the primary peaks is revealed by a careful study of the partial sums in calculating the electron density. In addition, when a bias is applied, the biggest change in wave function profile is at the corresponding resonant wave numbers $k_{\text{res}} \approx 4.8995 \times 10^8 \text{ m}^{-1}$. Accordingly, the cut-off wave number *must* be determined so that the primary peaks are included in the simulation domain. Moreover, the position and amplitude of the peaks are accurately modeled by a simplified model at zero applied voltage.

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